Momma's Bakeshop — Comprehensive Codebase Analysis

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1. PROJECT OVERVIEW

Field	Detail
Project Name	Momma's Bakeshop (`mommas-bakeshop`)
Description	A Point-of-Sale, Inventory, and Administration system for a bakeshop business
Project Type	Full-stack Web App + Desktop App (Electron wrapper)
Version	`0.1.0` (from package.json)
License	MIT (from composer.json)
Repository	Single-project (not a monorepo)

Repository Structure

```
IT12-Final-Project/
├── DBDump/                         # MySQL DB dump scripts & SQL backup files
│   ├── Backup.bat / Restore.bat     # Windows batch scripts for DB backup/restore
│   └── dbdump-mommas-bakeshop-data-only.sql
├── Screenshots/                    # System navigation screenshots & diagrams
├── IT12 ERD Final Project Draft.drawio  # Entity-Relationship Diagram
├── IT12 Final Project System Navigation.drawio
└── Mommas-Bakeshop/                # ★ Main application root
    ├── app/                        # Laravel backend (Models, Controllers, Middleware,
Services, Jobs)
    ├── bootstrap/                  # Laravel bootstrap
    ├── build/                      # Vite build output
    ├── config/                     # Laravel configuration files
    ├── database/                   # Migrations, seeders, factories
    ├── desktop/                    # Electron desktop app wrapper
    ├── docs/                       # Documentation
    ├── public/                     # Public web assets
    ├── references/                 # Reference materials
    ├── resources/                  # Frontend source (React/JSX, CSS, Blade views)
    │   ├── css/app.css
    │   ├── js/                     # React components, pages, layouts, hooks, utils
    │   └── views/app.blade.php     # Inertia root Blade template
    ├── routes/                     # Laravel routes (web.php, auth.php)
    ├── storage/                    # Laravel storage (logs, cache, sessions)
    ├── tests/                      # Pest/PHPUnit tests
    └── vendor/                     # Composer dependencies
```

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2. FRONTEND

Aspect	Technology	Version	Source File
Framework	React (via Inertia.js)	`^18.2.0`	package.json
SPA Adapter	Inertia.js for React	`^2.0.0`	`@inertiajs/react` in package.json
Language	JavaScript (JSX)	ES Modules	jsconfig.json
UI Component Library	Headless UI	`^2.0.0`	`@headlessui/react` in package.json
CSS / Styling	Tailwind CSS + `@tailwindcss/forms` + custom CSS variables	`^3.2.1`	tailwind.config.js, app.css
PostCSS	PostCSS + Autoprefixer	`^8.4.31` / `^10.4.12`	postcss.config.js
State Management	Inertia.js shared props (no dedicated state library)	—	HandleInertiaRequests.php
Routing	Server-side via Laravel + Ziggy (named route URLs in JS)	`^2.0`	composer.json (`tightenco/ziggy`)
Form Handling	Inertia.js `useForm` hook	—	Inline in page components
Data Fetching	Axios (for non-Inertia calls) + Inertia.js visits	`^1.11.0`	package.json
Icons	Lucide React	`^0.577.0`	package.json
Charts	Recharts	`^3.7.0`	package.json
PDF Generation	jsPDF + jspdf-autotable + html2canvas	`^4.2.0` / `^5.0.7` / `^1.4.1`	package.json
Theme Support	Light/Dark theme via `localStorage` + CSS class toggle	—	app.jsx
Custom Font	Figtree	—	tailwind.config.js

Not Found / Not Applicable

- **Validation Libraries** (Zod, Yup, etc.) — Not Found
- **Animation Libraries** — Not Found
- **Internationalization (i18n)** — Not Found
- **Rich Text Editor** — Not Found
- **Date/Time Libraries** — Not Found (uses native JS dates)
- **File Upload Handling** — Not Found (likely uses native form uploads)
- **SEO tools** — Not Applicable (admin/POS system, not public-facing)
- **PWA support** — Not Found
- **Accessibility (a11y) tools** — Not Found (Headless UI provides some built-in a11y)

Frontend Page Structure

```
resources/js/Pages/
├── Auth/                          # Login, Register, ForgotPassword, ResetPassword,
ConfirmPassword, VerifyEmail
├── Dashboard.jsx                  # Main dashboard with metrics
├── Welcome.jsx                   # Landing/welcome page
├── PointOfSale/
│   ├── CashSale.jsx              # Walk-in cash sales
│   ├── CashierTabs.jsx           # Cashier tab navigation
│   ├── Customers.jsx             # Customer management
│   ├── JobOrdersTabs.jsx         # Job orders module
│   ├── JobOrdersSubviews/        # Sub-components for job orders
│   ├── SaleHistoryTabs.jsx       # Sales history & pending payments
│   └── SaleHistorySubviews/      # Sub-components for sale history
├── Inventory/
│   ├── InventoryLevelsTabs.jsx   # Inventory levels, stock-in, stock-out, snapshots
│   ├── InventoryLevelsSubviews/  # Sub-components
│   ├── ProductsAndBatchesTabs.jsx# Products & production batches
│   ├── ProductsAndBatchesSubviews/
│   └── ShrinkageHistory.jsx      # Shrinkage tracking
├── Administration/
│   ├── Audits.jsx                # Audit trail viewer
│   ├── ReportsTabs.jsx           # Reports (Overview, Full Breakdown)
│   ├── ReportsSubviews/
│   ├── UserManagementTabs.jsx    # Users, Roles, Permissions, Permission Groups
│   ├── UserManagementSubviews/
│   ├── DatabaseTabs.jsx          # DB backups, connections, maintenance, schema, data
transfer, retention
│   └── DatabaseSubviews/
└── Profile/                      # User profile management
```

Components Library (15 components)

`ApplicationLogo` , `Checkbox` , `ConfirmationModal` , `DangerButton` , `Dropdown` , `InputError` , `InputLabel` , `Modal` , `NavLink` , `PrimaryButton` , `ResponsiveNavLink` , `SaleDocumentPreviewModal` , `SecondaryButton` , `TextInput` , `Toast`

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3. BACKEND

Aspect	Technology	Version	Source File
Framework	Laravel	`^12.0`	composer.json
Language	PHP	`^8.2`	composer.json
SPA Adapter	Inertia.js for Laravel	`^2.0`	`inertiajs/inertia-laravel`
API Architecture	Inertia.js (server-driven SPA, no traditional REST/GraphQL API)	—	web.php

Authentication	Laravel Breeze (session-based with Bcrypt hashing, 12 rounds)	`^2.3`	composer.json, .env
Authorization	Custom permission-based (RBAC with granular permissions)	—	EnsurePermission.php, User.php
ORM	Eloquent ORM (Laravel built-in)	—	All models in `app/Models/`
API Token Auth	Laravel Sanctum	`^4.0`	composer.json
Auditing	owen-it/laravel-auditing	`^14.0`	composer.json, audit.php
Queue	Laravel Queue (database driver)	—	.env (`QUEUE_CONNECTION=database`)
Cache	Laravel Cache (database driver)	—	.env (`CACHE_STORE=database`)
Session	Database-backed sessions	—	.env (`SESSION_DRIVER=database`)
Logging	Laravel Logging (stack → single channel)	—	.env, logging.php
Log Viewer	Laravel Pail	`^1.2.2`	composer.json
Mail	Laravel Mail (log driver for development)	—	.env (`MAIL_MAILER=log`)

Controllers (12 main + 9 Auth)

Controller	Responsibility
`DashboardController`	Dashboard metrics & widgets
`PosController`	Cash sales, job orders, sale history, customers, shrinkage (largest at ~45KB)
`InventoryController`	Inventory levels, stock-in/out, snapshots
`ProductController`	Products, production batches, categories, product snapshots
`CategoryController`	Product category CRUD
`ReportsController`	Reports (Overview, Full Breakdown)
`UserManagementController`	User CRUD, roles, permissions, permission groups
`AuditController`	Audit trail viewer
`DatabaseController`	DB backups, connections, migration, data transfer, schema report
`DesktopController`	Electron desktop health-check endpoint
`ProfileController`	User profile management
9 Auth controllers	Login, register, password reset, email verification (Laravel Breeze)

Middleware

Middleware	Purpose

`HandleInertiaRequests`	Shares auth user, permissions, flash messages, system state to frontend
`EnsurePermission`	Per-route permission enforcement (`permission:CanViewCashier`, etc.)
`EnsureLocalDesktopRequest`	Restricts desktop health endpoint to local requests
`EnsureNoMaintenanceWriteLock`	Blocks write operations during DB maintenance

Services

Service	Purpose
`DatabaseBackupService`	Snapshot & incremental backups, restore, verification, cleanup
`DatabaseConnectionManager`	Multi-database connection management (local ↔ remote switching)
`SystemOperationService`	Tracks ongoing system operations (maintenance locks)

Background Jobs (5)

`InitializeRemoteDatabaseJob`, `RestoreBackupJob`, `SwitchDatabaseTargetJob`, `TransferLocalToRemoteJob`, `VerifyBackupJob`

Not Found / Not Applicable

- **API Documentation** (Swagger/OpenAPI) — Not Found
- **Rate Limiting** — Not Found (beyond Laravel defaults)
- **File Storage / Upload** — Not Found
- **Email Service** — Log driver only (development)
- **Background Jobs framework** — Laravel Queue (database driver)
- **Caching framework** — Database driver only (no Redis)
- **Error Tracking** (Sentry, etc.) — Not Found
- **WebSocket / Real-time** — Not Found
- **Search Engine** — Not Found
- **Payment Processing** — Not Found (manual payment recording in POS)
- **Notification Services** — Not Found

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4. DATABASE

Aspect	Detail	Source
Primary Database	MySQL	.env (`DB_CONNECTION=mysql`)
Database Name	`mommas_bakeshop`	.env
Desktop DB Name	`mommas_bakeshop_desktop`	.env
Hosting	Self-hosted (localhost:3306)	.env
Migration Tool	Laravel Migrations	database/migrations/
Seeding	Laravel Seeders (5 seeders)	`database/seeders/`
Connection Pooling	Not Found (relies on Laravel defaults)	—

Secondary DB	`mysql_control` connection (for `system_operations` table)	Migration file

Database Schema (25+ tables)

```
erDiagram
    users ||--o{ permissions_set : "has"
    users ||--o{ audits : "generates"
    users ||--o{ sales : "processes"
    users ||--o{ shrinkages : "records"
    users ||--o{ stock_in_details : "creates"
    users ||--o{ stock_out_details : "creates"
    users ||--o{ production_batch_details : "creates"
    users }|--|| roles : "belongs to"

    roles ||--o{ role_preset_permissions : "has presets"
    permission_groups ||--o{ permissions : "groups"
    permissions ||--o{ permissions_set : "assigned via"
    permissions ||--o{ role_preset_permissions : "preset in"

    categories ||--o{ products : "contains"
    products ||--o{ production_batches : "produced in"
    products ||--o{ sold_products : "sold as"
    products ||--o{ shrinked_products : "shrinkage"
    products ||--o{ stock_ins : "stocked in"
    products ||--o{ stock_outs : "stocked out"
    products ||--o{ job_order_items : "ordered"
    products ||--o{ product_leftovers : "leftover in"

    sales ||--o{ sold_products : "contains"
    sales ||--o{ payments : "paid via"
    sales ||--o{ partial_payments : "partial"
    sales ||--o{ job_orders : "linked to"

    customers ||--o{ sales : "purchases"
    customers ||--o{ job_orders : "orders"

    job_orders ||--o{ job_order_items : "has items"
    job_orders ||--o{ job_order_custom_items : "has custom items"

    shrinkages ||--o{ shrinked_products : "contains"

    inventory ||--o{ stock_ins : "receives"
    inventory ||--o{ stock_outs : "dispatches"
    inventory ||--o{ inventory_leftovers : "snapshot"

    stock_in_details ||--o{ stock_ins : "contains"
    stock_out_details ||--o{ stock_outs : "contains"

    production_batch_details ||--o{ production_batches : "contains"
```

```
    product_leftover_snapshots ||--o{ product_leftovers : "records"
    inventory_leftover_snapshots ||--o{ inventory_leftovers : "records"

    database_backups ||--o{ database_backups : "base backup"
```

Key Tables

Table	Purpose
`users`	System users with role assignment
`roles`	User roles (Admin, Cashier, etc.)
`permissions`, `permissions_set`, `permission_groups`	Granular RBAC permission system
`role_preset_permissions`	Default permissions per role
`categories`, `products`	Product catalog with categories
`production_batches`, `production_batch_details`	Production tracking
`inventory`	Raw materials / supplies inventory
`stock_ins`, `stock_in_details`	Stock-in records with supplier info
`stock_outs`, `stock_out_details`	Stock-out records
`sales`, `sold_products`	Walk-in and job order sales
`payments`, `partial_payments`	Payment tracking (Paid/Partially Paid/Unpaid)
`customers`	Customer directory
`job_orders`, `job_order_items`, `job_order_custom_items`	Custom job orders
`shrinkages`, `shrinked_products`	Spoilage/theft/loss tracking with verification
`product_leftover_snapshots`, `product_leftovers`	End-of-day product inventory snapshots
`inventory_leftover_snapshots`, `inventory_leftovers`	End-of-day inventory snapshots
`audits`	Audit trail (Application + Trigger sources)
`database_backups`, `database_backup_changes`, `database_backup_settings`	Backup management system
`system_operations`	Maintenance lock tracking (on `mysql_control` connection)

Seeders

Seeder	Purpose
`DatabaseSeeder`	Orchestrates all seeders
`PermissionsSeeder`	Seeds all system permissions

`RolePermissionSeeder`	Seeds role-permission presets
`BakeshopCatalogSeeder`	Seeds sample products and categories
`ReportDemoSeeder`	Seeds demo data for reports

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5. DEVOPS, INFRASTRUCTURE & DEPLOYMENT

Aspect	Detail
Desktop Packaging	Electron `^40.8.0` + electron-builder `^26.8.1`
Build Output	`dist-electron/` (gitignored)
App ID	`com.mommasbakeshop.desktop`
Desktop Runtime	Bundled PHP + MySQL in `desktop/runtime/`
Containerization	Not Found (no Docker/Docker Compose files)
CI/CD Pipeline	Not Found (no `.github/workflows/` or CI config)
Infrastructure as Code	Not Found
Reverse Proxy	Not Found
CDN	Not Found
SSL/TLS	Not Found
Serverless	Not Found
Env Management	`.env` + `.env.example` (dotenv)
Monitoring	Not Found

Desktop App Architecture

The application can run as a standalone desktop app via Electron:

- main.mjs — Electron main process
- config.mjs — Desktop configuration
- process-manager.mjs — Manages PHP/MySQL child processes
- package.mjs — Build/packaging script
- `desktop/runtime/` — Bundled PHP & MySQL binaries

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6. TESTING

Aspect	Detail	Source
Unit/Feature Testing	Pest PHP `^3.8` + pestphp/pest-plugin-laravel `^3.2`	composer.json
Test Runner	PHPUnit (via Pest)	phpunit.xml

Mocking	Mockery `^1.6`	composer.json
Data Factories	Faker PHP `^1.23` + `UserFactory`	composer.json, `database/factories/`
Test Database	Uses `RefreshDatabase` trait against MySQL	Pest.php, phpunit.xml
Test Config	`phpunit.xml` with separate env vars (array cache, sync queue, etc.)	phpunit.xml

Not Found / Not Applicable

- **E2E Testing** (Cypress, Playwright) — Not Found
- **Frontend Testing** (Jest, Vitest, Testing Library) — Not Found
- **API Testing** — Not Found
- **Code Coverage** tooling — Not Found
- **Visual/Snapshot Testing** — Not Found
- **Load/Performance Testing** — Not Found

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7. CODE QUALITY & DEVELOPER EXPERIENCE

Aspect	Detail	Source
PHP Formatter	Laravel Pint `^1.24`	composer.json
Package Manager (PHP)	Composer	composer.json, `composer.lock`
Package Manager (JS)	npm	package-lock.json
Build Tool	Vite `^7.0.7` with `laravel-vite-plugin` `^2.0.0` + `@vitejs/plugin-react` `^4.2.0`	vite.config.js
Editor Config	`.editorconfig` (tabs, UTF-8, LF, final newline)	.editorconfig
Path Aliases	`@/*` → `resources/js/*`, `ziggy-js` → Ziggy vendor	jsconfig.json
Error Handling	nunomaduro/collision `^8.6` (pretty CLI error output)	composer.json
Concurrency	concurrently `^9.0.1` (runs server + queue + logs + vite in parallel)	composer.json `scripts.dev`

Not Found / Not Applicable

- **ESLint / JS Linter** — Not Found
- **Prettier / JS Formatter** — Not Found
- **TypeScript** — Not Found (uses plain JavaScript)
- **Git Hooks** (Husky, lint-staged) — Not Found
- **Commit Convention** — Not Found
- **Monorepo Tools** — Not Applicable
- **Documentation tools** (Storybook, Docusaurus) — Not Found

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8. THIRD-PARTY SERVICES & INTEGRATIONS

Aspect	Status
Analytics	Not Found
CMS	Not Found
Authentication Providers	Local email/password only
Cloud Storage	AWS S3 SDK configured but not active (.env keys empty)
Maps	Not Found
AI/ML Services	Not Found
Messaging / Chat	Not Found
Feature Flags	Not Found
A/B Testing	Not Found
Other APIs	Not Found

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9. SECURITY

Aspect	Detail	Source
Authentication	Session-based (Laravel Breeze) with Eloquent user provider	auth.php
Authorization	Custom RBAC: Users → Roles → Permissions, per-user permission overrides	`EnsurePermission` middleware, `permissions_set` table
Password Hashing	Bcrypt, 12 rounds	.env (`BCRYPT_ROUNDS=12`)
CSRF Protection	Laravel built-in (auto via Inertia)	—
Session Encryption	Disabled (`SESSION_ENCRYPT=false`)	.env
Password Timeout	3 hours (10800 seconds)	auth.php
Maintenance Lock	`EnsureNoMaintenanceWriteLock` middleware blocks writes during DB maintenance	Middleware
Desktop Local-Only	`EnsureLocalDesktopRequest` restricts desktop endpoints to localhost	Middleware
Audit Trail	Full audit logging (Application + Trigger-based)	`audits` table, `owen-it/laravel-auditing`

Not Found / Not Applicable

- **Security Headers** (Helmet.js, CSP) — Not Found
- **Rate Limiting** — Not Found (beyond Laravel defaults)
- **CORS Configuration** — Not Found (Inertia apps typically don't need CORS)
- **Input Sanitization / XSS** — Laravel auto-escaping
- **Security Scanning** (Snyk, Dependabot) — Not Found
- **Secrets Management** — `.env` files only

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10. CONFIGURATION FILES SUMMARY

File	Purpose
package.json	NPM dependencies, scripts, Electron build config
composer.json	PHP/Composer dependencies, autoload, scripts
.env	Environment configuration (DB, mail, cache, queue, desktop)
.env.example	Template for environment configuration
vite.config.js	Vite build config with Laravel + React plugins
tailwind.config.js	Tailwind CSS config with custom colors & font
postcss.config.js	PostCSS plugins (Tailwind + Autoprefixer)
jsconfig.json	JS path aliases for IDE resolution
phpunit.xml	PHPUnit/Pest test configuration
.editorconfig	Editor formatting rules
.gitignore	Git ignore rules
.gitattributes	Git attribute settings
`config/app.php`	Laravel app config (name, timezone, locale, providers)
`config/auth.php`	Authentication guards & providers
`config/audit.php`	Laravel Auditing configuration
`config/cache.php`	Cache store configuration
`config/database.php`	Database connections (mysql, mysql_control)
`config/database-backups.php`	Backup system settings
`config/database-connections.php`	Remote connection configuration
`config/desktop.php`	Electron desktop settings
`config/filesystems.php`	Filesystem disks
`config/logging.php`	Logging channels

`config/mail.php`	Mail transport settings
`config/queue.php`	Queue connection settings
`config/services.php`	Third-party service credentials
`config/session.php`	Session configuration

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11. DEPENDENCY ANALYSIS

Production Dependencies (npm: 5, composer: 7)

Package	Version	Purpose
`html2canvas`	`^1.4.1`	Screen capture for PDF generation
`jspdf`	`^4.2.0`	PDF document creation
`jspdf-autotable`	`^5.0.7`	Table generation in PDFs
`lucide-react`	`^0.577.0`	Icon library
`recharts`	`^3.7.0`	Chart/data visualization
`laravel/framework`	`^12.0`	Core PHP framework
`laravel/sanctum`	`^4.0`	API token authentication
`laravel/tinker`	`^2.10.1`	REPL for Laravel
`inertiajs/inertia-laravel`	`^2.0`	Server-side Inertia adapter
`owen-it/laravel-auditing`	`^14.0`	Model audit trails
`tightenco/ziggy`	`^2.0`	Named route URLs in JavaScript

Dev Dependencies (npm: 12, composer: 7)

Package	Version	Purpose
`react` / `react-dom`	`^18.2.0`	UI framework
`@inertiajs/react`	`^2.0.0`	Client-side Inertia adapter
`@headlessui/react`	`^2.0.0`	Accessible UI components
`tailwindcss`	`^3.2.1`	Utility-first CSS
`@tailwindcss/forms`	`^0.5.3`	Form styling plugin
`@tailwindcss/vite`	`^4.0.0`	Tailwind Vite integration
`vite`	`^7.0.7`	Build tool
`@vitejs/plugin-react`	`^4.2.0`	React plugin for Vite

`laravel-vite-plugin`	`^2.0.0`	Laravel plugin for Vite
`autoprefixer`	`^10.4.12`	CSS vendor prefixing
`postcss`	`^8.4.31`	CSS processing
`axios`	`^1.11.0`	HTTP client
`concurrently`	`^9.0.1`	Parallel process runner
`electron`	`^40.8.0`	Desktop runtime
`electron-builder`	`^26.8.1`	Desktop packaging
`laravel/breeze`	`^2.3`	Auth scaffolding
`laravel/pail`	`^1.2.2`	Real-time log viewer
`laravel/pint`	`^1.24`	PHP code formatter
`laravel/sail`	`^1.41`	Docker dev environment (not actively used)
`mockery/mockery`	`^1.6`	PHP mocking library
`nunomaduro/collision`	`^8.6`	Pretty error reporting
`pestphp/pest`	`^3.8`	PHP testing framework
`pestphp/pest-plugin-laravel`	`^3.2`	Pest plugin for Laravel
`fakerphp/faker`	`^1.23`	Test data generation

> *[!NOTE] **Potential Version Mismatch**: `@tailwindcss/vite` is at `^4.0.0` while `tailwindcss` is at `^3.2.1`. The `@tailwindcss/vite` v4 plugin is designed for Tailwind CSS v4, but the project uses v3's config format.*

Totals

- **npm production**: 5 packages
- **npm dev**: 12 packages
- **Composer production**: 7 packages
- **Composer dev**: 7 packages

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12. ARCHITECTURE PATTERNS & OBSERVATIONS

Architecture Pattern

- **MVC** (Model–View–Controller) via Laravel conventions
- **Server-Driven SPA**: Inertia.js bridges server-side routing with React client rendering
- **Hybrid Web/Desktop**: Same codebase serves both web and Electron desktop deployments

Design Patterns Observed

- **Repository-like Service Layer**: `DatabaseBackupService`, `DatabaseConnectionManager`, `SystemOperationService` encapsulate complex business logic
- **Middleware Pipeline**: Authentication → Permission checking → Maintenance lock → Controller

- **Observer/Audit Pattern**: Automatic audit trail generation via `owen-it/laravel-auditing` + database triggers
- **Factory Pattern**: `UserFactory` for test data generation
- **RBAC with Per-User Overrides**: Roles define preset permissions; individual users can have permission overrides via `permissions_set`

Code Organization

- **Layer-Based**: Standard Laravel structure (`app/Models` , `app/Http/Controllers` , `app/Services` , `app/Jobs`)
- **Feature-Based Frontend**: React pages organized by module (`PointOfSale/` , `Inventory/` , `Administration/`)
- **Tab-Based Navigation**: Many pages use tab sub-views (e.g., `InventoryLevelsTabs.jsx` with `InventoryLevelsSubviews/`)

Error Handling

- Permission errors return 403 (JSON, Inertia redirect, or abort)
- Flash messages for success/error feedback (`session('success')` , `session('error')`)
- Maintenance lock returns write-blocked errors

Notable Conventions

- PascalCase column names in DB (e.g., `CustomerName` , `ProductID`) — non-standard for Laravel
- Custom `ID` primary keys instead of Laravel's default `id`
- Database triggers used alongside application-level auditing
- CSS custom properties for theme colors (`--color-primary` , etc.)
- Dark/Light theme toggle persisted in `localStorage`

Potential Technical Debt / Concerns

> *[!WARNING]*
>
> - ***`PosController` is very large*** *(~45KB) — handles cash sales, job orders, shrinkage, and customers. Should be split.*
> - ***`UserManagementController`*** *(~~32KB) and~~* ***~~`ReportsController`~~*** *~~-~~(25KB) are also oversized.*
> - ***No frontend linting*** *(ESLint/Prettier) — code style enforcement is missing for JS/JSX.*
> - ***No frontend tests*** *— only Pest PHP tests exist; no Jest/Vitest/Cypress.*
> - ***Session encryption disabled*** *— `SESSION_ENCRYPT=false` in production could be a concern.*
> - ***`.env` committed to repo*** *— should typically be gitignored (it IS in `.gitignore` but exists).*
> - ***Tailwind v3/v4 mismatch*** *— `@tailwindcss/vite` v4 with Tailwind v3 config may cause issues.*
> - ***No CI/CD*** *pipeline defined.*
> - ***No Docker*** *setup despite `laravel/sail` being in dev dependencies.*

Scalability Considerations

- Database-backed sessions, cache, and queue — fine for single-server but won't scale horizontally without migration to Redis
- Desktop app bundles PHP + MySQL runtime — creates large distributable but enables offline operation
- Multi-database connection management suggests planned remote DB deployment

Missing Best Practices Recommendations

1. Add ESLint + Prettier for frontend code quality
2. Add frontend unit/integration tests (Jest/Vitest + Testing Library)

3. Add E2E tests (Cypress or Playwright)
4. Split large controllers into smaller, focused controllers or use Action classes
5. Set up CI/CD (GitHub Actions)
6. Enable session encryption for production
7. Resolve Tailwind v3/v4 package mismatch
8. Add rate limiting to authentication routes
9. Add security headers (CSP, HSTS) for production deployment

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SUMMARY TABLE

Category	Technology	Version	Notes
Frontend Framework	React	^18.2.0	Via Inertia.js
SPA Adapter	Inertia.js	^2.0.0	Server-driven SPA
UI Library	Headless UI	^2.0.0	Accessible components
CSS	Tailwind CSS	^3.2.1	+ @tailwindcss/forms
Icons	Lucide React	^0.577.0	—
Charts	Recharts	^3.7.0	—
PDF	jsPDF + autotable	^4.2.0	+ html2canvas
Backend Framework	Laravel	^12.0	PHP ^8.2
Auth Scaffolding	Laravel Breeze	^2.3	Session-based
API Token Auth	Laravel Sanctum	^4.0	—
Auditing	laravel-auditing	^14.0	+ DB triggers
Route URLs (JS)	Ziggy	^2.0	Named routes in React
Database	MySQL	localhost:3306	`mommas_bakeshop`
ORM	Eloquent	Built-in	36 models
Migration Tool	Laravel Migrations	Built-in	5 migration files
Test Framework	Pest PHP	^3.8	+ Mockery, Faker
Build Tool	Vite	^7.0.7	+ laravel-vite-plugin
PHP Formatter	Laravel Pint	^1.24	—
Desktop Runtime	Electron	^40.8.0	+ electron-builder
Package Manager	npm + Composer	—	—
Queue/Cache/Session	Database driver	—	All database-backed